## Daniel Esteban Cárdenas Jerez

AI & Systems Engineer — Applied AI, Backend & Edge ML

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### Profile

Engineering Physics graduate building practical AI systems across documents, voice workflows, operational data and connected devices. Combines Python/backend engineering with retrieval, evaluation, computer vision and embedded Linux. Turns model output into reviewable decisions through explicit state, authorization, provenance and failure handling.

### Experience

**Lean Tech — AI Engineer** · Remote · Feb 2026 – Present

- Lead architecture, scope and release acceptance for a production Teams document assistant delivering editable SOPs, flowcharts and presentations; coordinate AI-assisted implementation of requester isolation, source tracking, revisions and consent-aware delivery.
- Own reliability and evaluation improvements for an inherited, collaboratively developed Spanish-language voice platform. Retrospective cohorts showed **~70.1% lower mean recorded cost among costed calls** ($0.150631 → $0.045053); missing billing and cohort differences limit causal interpretation.
- Design reproducible prompt evaluations, separate confirmed from ambiguous failures, and preserve inconclusive comparisons rather than claim model superiority.
- Advise on CRM integration and contribute focused identity/delivery fixes; extend document and operational workflows with Python, TypeScript, APIs, SQL and human-review controls.

**Solenium — ML Engineer / Firmware Developer** · Medellín · Jan 2025 – Feb 2026

- Led the ML component of a smart-meter energy-disaggregation platform, integrating inference with embedded Linux/Yocto, telemetry, retraining and device recovery; historically reported operation at **10 Hz with sub-2-second inference**.
- Developed field-attendance computer vision with face matching, liveness, OCR, offline handling and API integration; supported physical-device debugging and operational handover.
- Championed AI coding-tool adoption and helped organize cross-functional prototyping workshops with technical and business stakeholders.

**XM S.A. E.S.P. — Market Development Analyst** · Medellín · Jul 2024 – Jan 2025

- Built Python/pandas reconciliation across databases, files and APIs, producing auditable reports and resolving operational discrepancies with stakeholders.

**Magnus — Machine Learning Engineer (Full-time)** · Remote · May 2023 – Jul 2024

- Built Azure OpenAI applications for clinical simulations, medical training and survey feedback, with prompt testing/evaluation and RAG retrieval workflows.

### Selected Projects

- **Multilingual RAG:** FastAPI, BGE-M3, pgvector, local models and source citations; retrieval and latency evaluation.
- **Backend automation:** idempotent Python/SQLite ledger reconciliation, effective-dated policies, atomic writes, deterministic payment keys and regression checks.
- **Agent/platform operations:** directed personal browser and messaging controls with identity rechecks, bounded queues and uncertainty-aware delivery; coordinated security reviews and Linux incident diagnosis with agent collaborators.
- **Connected systems:** ESP32/Azure MQTT/TLS telemetry, camera/OCR monitoring and confirmed voice-to-inventory workflows.

### Technical Skills

**AI & data:** Python, RAG, agents, Azure OpenAI, Gemini, LangGraph, evaluation, guardrails, OCR, PostgreSQL/pgvector, FAISS, Pinecone, SQLite, pandas

**Product & platform:** FastAPI, Flask, TypeScript, React, Supabase, REST, MCP, Docker, CI/CD, Azure/AWS, Linux, audit trails, idempotency, access controls

**Embedded:** C, STM32, FreeRTOS, ESP32, Yocto, MQTT, telemetry, OTA/recovery

### Education & Languages

B.Sc. Engineering Physics — Universidad Nacional de Colombia, 2025 · IBM AI/Deep Learning and Generative AI certificates · AWS Academy Cloud Foundations

English — C1 (IELTS 7.0) · Spanish — Native · Portuguese — A2–B1 (self-assessed; not certified)